

Improved relation extraction through key phrase identification using community detection on dependency trees

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ABSTRACT

A method for extracting relations from sentences by utilizing their dependency trees to identify key phrases is presented in this paper. Dependency trees are commonly used in natural language processing to represent the grammatical structure of a sentence, and this approach builds upon this representation to extract meaningful relations between phrases. Identifying key phrases is crucial in relation extraction as they often indicate the entities and actions involved in a relation. The method uses community detection algorithms on the dependency tree to identify groups of related words that form key phrases, such as subject-verb-object structures. The experiments on the Semeval-2010 task8 dataset and the TACRED dataset demonstrate that the proposed method outperforms existing baseline methods.

1. Introduction

Relation extraction task aims to detect the relationship between two given entities in the input text. The triplets extracted from the text are widely used in many downstream natural language processing tasks (NLP), such as question-answering systems (Xu et al., 2016), knowledge graph construction (Ye et al., 2022).

Models making use of dependency parses of the input sentences have proven to be very effective in relation extraction. Existing models for utilizing the syntactic structure information provided by dependency trees can be divided into those using LSTM (Tai et al., 2015; Xu et al., 2015; Zhang et al., 2017) and those using GCN. In recent years, most models have utilized graph convolutional networks for dependency tree modeling due to its parallel computing capability, and develop some pruning strategies to filter out noisy information that confuses the model. However, the use of subtree-based hard pruning rules in some models (Zhang et al., 2018) result in the loss of important information. To address this problem, (Guo et al., 2019) proposes a soft pruning strategy based on the multi-head attention mechanism.

Despite these advancements, sentence-level relationship extraction methods still rely on subtree-based hard pruning or soft pruning strategies that primarily target individual words. This approach overlooks the segment-level features, the importance of which has been demonstrated in previous works in the field of information extraction. Additionally, the use of convolutional neural network structure (Liang et al., 2022; Wang et al., 2016) for key phrase extraction has notable limitations: (1) the convolutional kernel size is predetermined and fixed, hence it fails to effectively handle multi-word phrases of varying lengths in the text. (2) convolutional neural networks can only recognize and extract phrases composed of consecutive words. These limitations restrict the performance of

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convolutional neural networks in sentence-level relation extraction task.

We employ a simple example in Fig. 1 to demonstrate the potential issues caused by the limitations of convolutional neural network, and to introduce the solution method we propose. Given the sentence and entities pair ("quarterback", "pass"), we can find that the relation "*Instrument-agency*" of given entity "quarterback" and entity "pass" is built upon the core phrase "*wins the game*" and "*with a pass*". However, because the key phrase "*with a pass*" is separated by the proper noun "*Hail Mary*", using a traditional convolutional neural network structure for key-phrase identification may not be effective. It is worth noting that although the key phrase "*with a pass*" is not continuous in the original sentence, it is tightly connected in the dependency tree.

This work proposes a novel approach to improve relation extraction by identifying phrases through community detection on dependency trees. Community detection groups related words in a sentence based on their connectivity patterns, and we treat each group as a phrase. This approach allows for more accurate identification of key phrases. The corresponding dependency tree partitioning result for the example is shown in the Fig. 2. We can observe that even non-continuous key phrases such as "*with a pass*" can be identified by analyzing the dependency tree. Our approach also considers co-reference of entities and descriptive information of entities that were less focused in previous work, such as "*wins the game*" being the descriptive information for entity "quarterback". The main message of the sentence is then selected based on a high correlation coefficient with the two entities, while ignoring noisy information such as proper noun "*Hail Mary*", that may confuse the model. Our method overcomes the limitations of existing methods that rely on pruning strategies to filter out noise information, and achieves better performance in relation extraction tasks.

Our contributions are summarized as follows:

- We proposed a novel approach that combines dependency trees and community detection algorithms to identify phrases in sentences based on their topological structure. Experimental results demonstrate the effectiveness of this approach.
- The model introduces a hierarchical attention mechanism that enhances entity representations by incorporating entity-context information, allowing the generated entity representations to capture global semantic information related specifically to the entities.
- Experiments conducted on the SemEval-2010 task 8 dataset and the TACRED dataset show that our proposed model achieves state-of-the-art results among Pipeline-based models.

2. Related work

Relation extraction is a fundamental task in natural language processing, and various methods have been proposed to extract relations between entities from unstructured text. Traditional rule-based approaches heavily rely on manually designed rules, which are time-consuming and inefficient. To address this issue, statistical methods Tree-based kernels (Zelenko et al., 2002) and dependency path-based kernels (Bunescu and Mooney, 2005) are explored to extract the relation. While traditional machine learning methods are more general and efficient than rule-based methods, designing features and kernel functions is more challenging and laborious.

Recent research has shown that neural networks can achieve better performance and higher efficiency than traditional feature-based methods, as they can automatically learn rich and expressive feature representations from data. These models introduce various paradigms to tackle the relation extraction task.

The most typical training frame for relation extraction models is to treat the task as a multi-classification task, where the goal is to predict the relation label for each pair of entities in the input text. Based on the steps taken by the model to solve the task, classification-based methods can be divided into two forms: pipeline and joint entity and relation extraction. The joint entity and relation extraction models (Ye et al., 2022; Zheng et al., 2017) simultaneously optimize entity recognition and relation extraction. This joint optimization aims to leverage the interaction between these two tasks, thereby enhancing overall performance. The pipeline models consist of separate stages for entity recognition and relation extraction, offering greater flexibility. Benefiting from the advancement of PLMs, the pipeline models have attained impressive performance in RE tasks by incorporating additional components onto PLMs. To leverage dependency tree information, A-GCN (Tian et al., 2021) employ BERT (Devlin et al., 2018) as the encoder and concatenate multiple layers of GCN to model the dependency tree. Additionally, the model consider the dependency relations between words to assign weights to each edge. SMS (Liang et al., 2022) utilizes CNN to extract segment-level features, thus enhancing the model performance.

In recent research, there has also been some attention drawn to prompt-based methods. PTR (Han et al., 2022) and KnowPrompt (Chen et al., 2022) employ a manually crafted prompt along with distinct label verbalizers for individual relations. GenPT (Han et al., 2022) breaks the fixed word count limitation of verbalizers by using a generative model to generate relation labels.

Using sequence-to-sequence models for relation extraction tasks is another research hotspot. REBEL (Huguet Cabot and Navigli, 2021) pretrains the BART-large (Lewis et al., 2020) model using external datasets to enhance the model's performance in information extraction tasks. TANL (Paolini et al., 2021) utilizes the T5 generation model as the backbone network, transforming structured prediction tasks into translation tasks to leverage the potential knowledge about labels inherent in PLMs.



Fig. 1. An example of relation extraction. The underlined words denote the given entities, and the bold phrases are the key information.

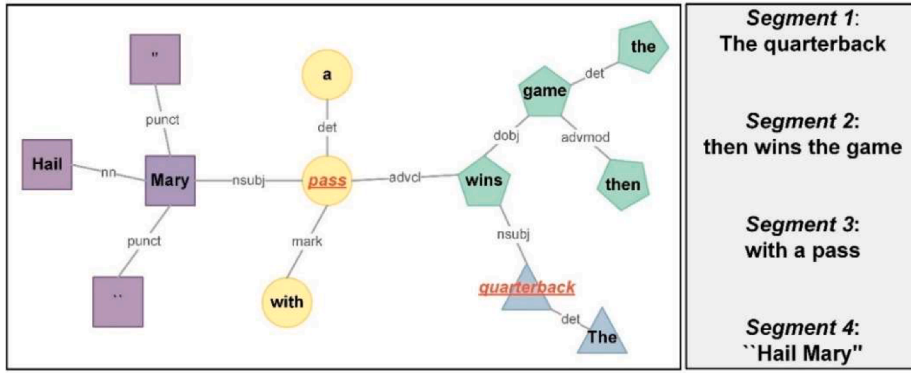


Fig. 2. Partition result for the corresponding example. Nodes (words) with the same shape are grouped together into the same segment.

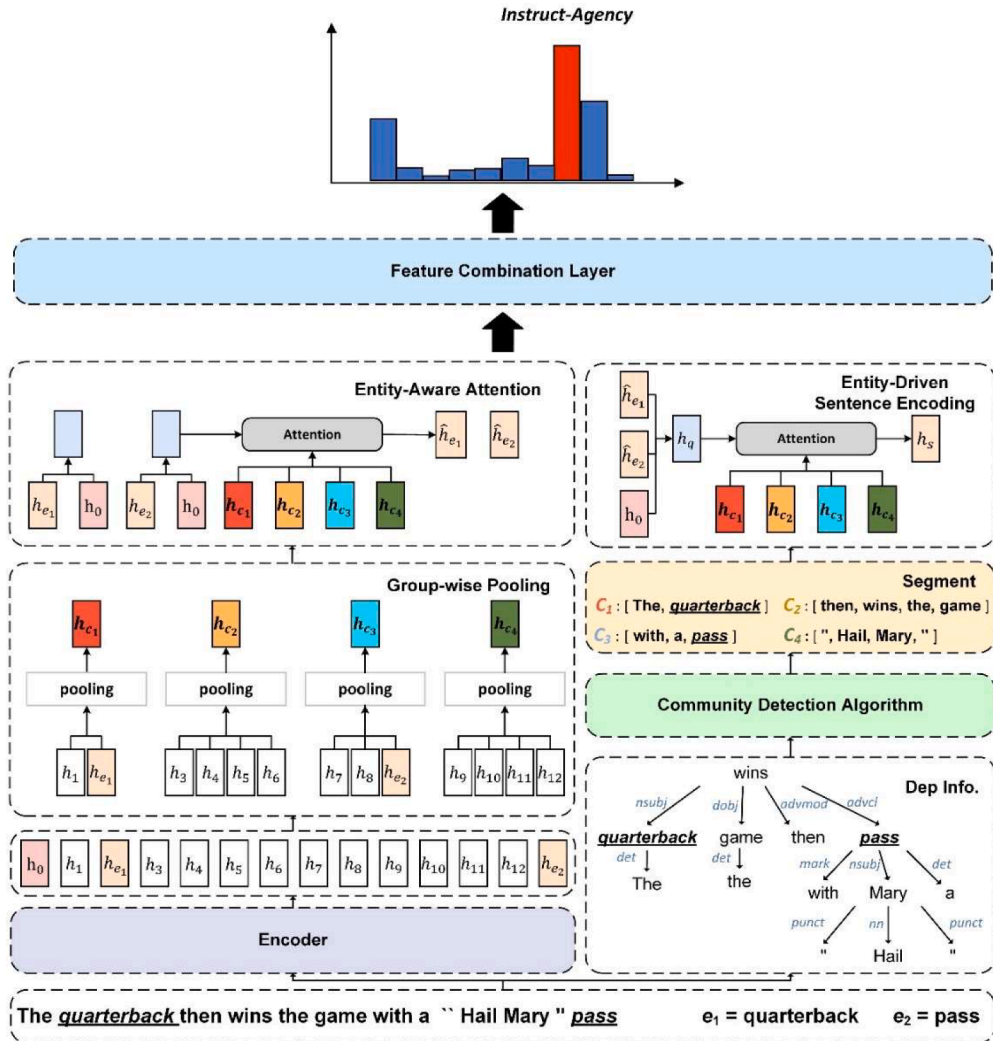


Fig. 3. The overall architecture of the proposed model illustrated with an example input sentence (the two entities are marked by underscores) and its dependency tree.

3. Proposed method

In this section, we will provide a detailed explanation of the fundamental components that make up the BCommDet model. Fig. 3 shows the structure of our model.

3.1. Encoder layer

Given a sentence $S = \{w_1, \dots, w_n\}$ and corresponding subject and object entities identified by the spans $e_1 = \{w_{s_1}, \dots, w_{s_2}\}$ and $e_2 = \{w_{o_1}, \dots, w_{o_2}\}$, It is necessary to preprocess the sentence before it can be fed into a text encoder (e.g. BERT (Devlin et al., 2018), RoBERTa (Liu et al., 2019)). Specifically, two special tokens are added to the beginning and end of the sentence respectively. After the preprocessing step, a text encoder is used to map the tokens in the input sentence to vector representations. The resulting vector representation is denoted as:

$$H = \text{encoder}(w_0, \dots, w_{n+1}) \quad (1)$$

where $H = \{h_0, \dots, h_n\}$ is the vector representation of the input sentence. Notably, h_0 represents the feature representation of starting token of the sentence and contains the semantic information of the entire sentence.

3.2. Community detection algorithm

We used the Leiden algorithm (Traag et al., 2019) to perform community detection on the dependency tree of each sentence. The Leiden algorithm is a widely used modularity-based community detection algorithm that can efficiently identify clusters or communities in graphs. Compared to other community detection algorithms based on different principles, the Leiden algorithm offers the following advantages in our application scenario: (1) Modularity-based algorithms provide an intuitive and easily interpretable explanation for a partition result, aiding in better understanding the organizational structure of networks. (2)

Due to variations in sentence syntax, it's challenging to determine the number of communities per sentence. Leiden algorithm's adaptive approach eliminates the need for users to specify community numbers in advance.

The process of the Leiden algorithm involves optimizing the modularity score of the graph partition, which measures the quality of the partition based on the density of edges within communities and the sparsity of edges between communities. A higher modularity score indicates a stronger community structure. Specifically, the modularity score is calculated using the formula:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{(2m)} \right] \delta(c_i, c_j) \quad (2)$$

Where m is the total number of edges in the graph, A_{ij} is the weight of the edge between nodes i and j , k_i and k_j are the degrees of nodes i and j , c_i and c_j are the community assignments of nodes i and j , and δ is the Kronecker delta function, the function evaluates to 1 when nodes i and j are assigned to the same community c , and 0 otherwise. The Leiden algorithm has a time complexity of $O(kn \log n)$, where k is the number of iterations and n is the number of nodes in the graph. However, in practice, the algorithm often converges after a few iterations, leading to a practical time complexity that is much lower than the worst-case time complexity.

To apply the Leiden algorithm to our task, we first converted the dependency tree of each sentence into an undirected graph. Given a sentence with n words, we represent each word as a node w_i , where i is the index of the word in the original sentence, and the dependency relation between words w_i and w_j was represented as an edge, denoted as (w_i, w_j) . We assigned the same weight value of 1 to each edge.

The resulting undirected graph is defined by the point set V and edge set E , where $V = \{w_1, \dots, w_n\}$ and $E = \{(w_i, w_j) \mid \text{If there exists a dependency between words } w_i \text{ and } w_j\}$. We represent the graph of the sentence as $G = (V, E)$ and use the Leiden algorithm to partition it into communities of words that are more closely related to each other than to words in other communities. The Leiden algorithm returns a set of communities $C = \{C_1, C_2, \dots, C_t\}$ that maximizes the modularity Q , where t is the number of communities.

Each community C_i is defined as a subset of the node set V in the graph G , expressed as $C_i = \{w_j \mid w_j \in V \text{ and } w_j \text{ belongs to community } i\}$. The resulting communities correspond to phrases in the sentence and provide useful information for relation extraction.

3.3. Group-wise pooling

Upon applying the community detection algorithm to the dependency tree corresponding to a sentence, we obtain a partition that corresponds to the sentence. To obtain the feature expression for each phrase before selecting them with an attention mechanism, we use the following formula:

$$h_{c_i} = W \left[\tanh \left(\frac{1}{|C_i|} \sum_{w_j \in C_i} h_j \right) \right] + b \quad (3)$$

Here, h_{c_i} represents the feature expression for the phrase that corresponds to community C_i . W and b are the weight matrix and bias

term, respectively, and h_j is the feature vector for word w_j belonging to community C_i . By utilizing the activation function, this formula compresses the average of the feature vectors into a fixed length vector, which captures the overall characteristics and patterns of the community.

3.4. Entity-aware attention

After obtaining the feature vectors corresponding to each community, it becomes essential to filter and extract only the relevant information related to the two entities of interest. In this crucial step, the primary objective is to extract the entity context information from the sentence, such as entity's action or co-reference. Given the high relevance of this information to the entities, we can leverage the attention mechanism (Vaswani et al., 2017) to extract it. It is worth noting that some descriptive information of entities in sentences, such as "adj-and-adj" phrases, is highly relevant to entities but adds redundancy when determining the relationship between entities. Thus, it is crucial to ignore this information and capture only the key information. This goal can be achieved by utilizing the sentence backbone information contained in the [CLS] token. Specifically, the embeddings of the CLS token and the entity are concatenated and passed through a parameter matrix as a query vector to obtain the entity context features from $H_c = \{h_{c_1}, \dots, h_{c_t}\}$ by Equ.(5).

$$\hat{h}_{e_1} = \text{Softmax}\left(\frac{H_c \cdot W_e[h_{e_1}; h_0]}{\sqrt{d}}\right) \cdot H_c + h_{e_1} \quad (4)$$

Where $W_s \in R^{d \times 2d}$ is a linear transform matrix, h_{e_1} is the embedding representation of the subject, and d is a hidden dimension of vectors. By incorporating information from entity context, we create a more robust and informative representation of entity. Similarly, the feature representation of the object entity feature $\hat{h}_{e_2}$ can be obtained in the same way.

3.5. Entity-driven sentence encoding

The Entity-driven sentence encoding is a hierarchical structure that utilizes entity mention features $\hat{h}_{e_1}$ and $\hat{h}_{e_2}$ from Entity-aware attention. Although we have captured contextual information about the given entities in the previous step, it ignores the information in the sentence that only relates to the two entities to form the sentence backbone, which can assist the model in predicting the relationships between entities in instances that lack explicit patterns of phrases.

To obtain information about the sentence backbone, we use an attention operation called Entity-driven sentence encoding, which employs the concatenation of $[\hat{h}_{e_1}; \hat{h}_{e_2}; h_0]$ as a query to capture backbone features from H using Equ. (6). By obtaining the embedding of the whole sentence in this way, the semantic information it contains is related to only two entities, thus ignoring irrelevant and interfering information in the sentence, and ultimately improving the model performance.

$$h_s = \text{Softmax}\left(\frac{H_c \cdot W_s[\hat{h}_{e_1}; \hat{h}_{e_2}; h_0]}{\sqrt{d}}\right) \cdot H_c \quad (6)$$

Where $W_s \in R^{d \times 3d}$ is a linear transform matrix, h_s is the sentence embedding.

3.6. Feature combination layer and relation classification

The feature combination layer aggregates the global semantic information and entity features obtained from previous modules to form the classification feature inputs for the classifier. The specific formula used to accomplish this is as follows:

$$h_g = \text{ReLU}(W_a[h_s; \hat{h}_{e_1}; \hat{h}_{e_2}]) \quad (7)$$

In this formula, h_g is the output feature vector of the combination layer, W_a is a weight matrix, h_s is the output feature vector of the previous layer, and $\hat{h}_{e_1}$ and $\hat{h}_{e_2}$ are the enhanced feature expression of two entities. The ReLU function is used to introduce nonlinearity into the output. We use a softmax function to generate the probability of each relation label, which is represented by the Equ.(8):

$$P(r|S, e_1, e_2) = \text{Softmax}(W_o h_g) \quad (8)$$

Then, we train our model using the cross-entropy loss function. We refer to our model as CommDet.

4. Experiment

4.1. Dataset and evaluation metrics

To evaluate the proposed model in this paper, we conducted training and testing on two datasets: SemEval-2010 Task8 and TACRED. The SemEval-2010 Task 8 dataset consists of 8000 training samples and 2717 testing samples. This dataset includes nine specific relation types along with a negative category labeled as "Other". The TACRED dataset comprises 68,124 training samples, 22,631 validation samples, and 15,509 testing samples. It introduces 41 relation types along with a special "no relation" type to describe relations between mention pairs in samples. Consistent with prior work, we use Macro-F1 score excluding 'No relation' to

evaluate model performance. All reported results are the mean of five results with different randomly selected seeds.

4.2. Settings

The hyperparameters for the proposed model are listed in Table 1. We employ the “entity marker” strategy to emphasize the position of entities within sentences. Specifically, for the SemEval-2010 Task8 dataset, we add the token [E1] before the subject and the token [/E1] at the end of the subject. We also apply the same approach to the object and use the token [E2] and [/E2]. For the TACRED dataset, we use entity types as auxiliary information, and we employ the “Typed entity marker(punct)” strategy proposed by (Zhou and Chen, 2021). We represent the hidden representation h_{e_i} of the subject using the embedding representation of the subject’s starting token (e.g. [E1]) obtained from the encoder output, and the hidden representation of the object is obtained in the same manner. During the Group-wise pooling stage, these added special tokens are removed from the sequence to ensure alignment between the sentence and the dependency tree structure. We use the We employed the Leiden method from the igraph library, which is an open-source library for creating and analyzing graphs and networks, to partition the dependency tree.

All experiments in this study utilized RoBERTa-large as the encoder. To ensure seamless integration with RoBERTa, the output dimension of each module in our model was all set to match that of RoBERTa.

4.3. Experiment result

The proposed method employs a community detection algorithm to partition the dependency tree of each sentence, which in turn facilitates the extraction of key phrases for aiding in the task of relation extraction between entities. In this section, we present the experimental results of our method in Tabel 2 and compare its performance with SOTA models in the field, including Classification-based model, Sequence-to-sequence model, Prompt-based model.

Table 2 shows that our model achieved an F1 score of 90.8 % on the SemEval 2010 task-8 dataset and 75.6% on the TACRED dataset, significantly outperforming existing state-of-the-art models. The training tasks of Prompt-based models are more closely aligned with pretraining tasks, allowing them to fully leverage the semantic knowledge contained in the model. However, due to the need to design verbalizations and corresponding prompts for specific relations, they lack some generality. Additionally, SMS utilizes CNN to extract segment-level features, and its efficient performance demonstrates the importance of key phrases in relation extraction tasks. Notably, our proposed relation extraction model exhibits significant performance enhancement in this task, demonstrating its effectiveness in accurately extracting relations between entities.

5. Analysis and discussion

5.1. Effectiveness analysis of community detection

Because the structure of a sentence’s dependency tree can vary greatly depending on its length, we need to first test the performance of the community detection algorithm on dependency trees of different sizes. Specifically, we divide the training set into multiple groups based on sentence length, and evaluate the quality of the partition results by calculating the modularity of each group. The results are shown in the Fig. 4.

From the figure, we observe that as the length of the sentence increases, the modularity of the partition also increases. This is because as the size of the tree increases, the probability of forming dense clusters with many internal connections and fewer connections to nodes in other clusters also increases, resulting in higher modularity values. A modularity value ranging from 1 to 1, with a value between 0.3 and 0.8 generally considered to indicate a strong community structure within a network partition. With most modularity values above 0.3 for partition results of different sentence lengths, it suggests that there is a strong community structure within each partition of the network created by the dependency tree.

To further investigate whether the community detection algorithm can effectively identify key phrase structures from the original sentences and positively impact the model, we conducted comparative experiments using different methods to obtain features as inputs for the Entity-aware attention (EA) and Entity-driven sentence encoding (EDS) modules, and the results are presented in Table 3.

In the One Token model, we use individual token embeddings as input features. For the CNN model, we utilize various kernel sizes of 1, 3 and 5 to capture phrases at different levels of granularity, and the resulting feature maps are max-pooled to represent phrases with the central word as the focal point.

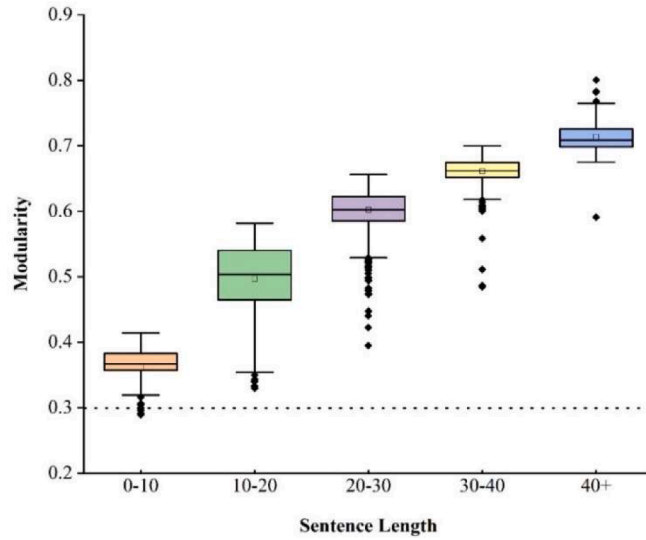
Table 1
Hyper-parameter of model.

Hyper-parameter	Value
Max sequence length	256
Batch Size	32
Initial Learning Rate for Adam	5e-5
Number of Training Epoch	5.0
Droupout Rate	0.1
Warmup Ratio	0.2

Table 2

Micro-F1 scores of test sets on the SemEval-2010 Task 8 dataset and the TACRED dataset. * means that the results are obtained from (Li et al., 2022). † means that the model employs dependency tree information.

Model class	Model	TACRED	SemEval
Classification-based model	AGCN†(Tian et al., 2021)	–	89.85
	SMS(Liang et al., 2022)	69.9	90.3
	LUKE(Yamada et al., 2020)	72.7	90.1
Sequence-to-sequence model	BART*(Lewis et al., 2020)	72.7	–
	REBEL*(Huguet Cabot and Navigli, 2021)	73.7	–
	TANL*(Paolini et al., 2021)	74.8	–
Prompt-based model	PTR(Han et al., 2022)	72.4	89.9
	KnowPrompt (Chen et al., 2022)	72.4	90.2
	GenPT(Han et al., 2022)	75.3	–
Our Model	CommDet	75.6	90.8

**Fig. 4.** The trend in modularity changes as sentence length increases.**Table 3**

Comparative results (F1-score) of different feature extraction methods.

Model	SemEval
One Token	90.1
CNN	90.3
CommDet	90.8

Our experiments revealed that using a CNN structure to capture key phrases was more effective than relying solely on individual tokens. This demonstrating the significant role that key phrases play in identifying the relationship between entities. Additionally, our model performs better than both models using different feature extraction methods, it indicated that utilizing the community detection algorithm to identify phrases and extract key phrases was more effective than using a CNN structure alone.

Table 4

Comparative results (F1-score) of different components.

Model	TACRED	SemEval
CommDet	75.6	90.8
- EA	74.9	90.6
- EDS	74.6	90.3

5.2. Effectiveness analysis of different components

To evaluate the contribution of each component of our model, we do an ablation study on the Semeval-2010 task 8 dataset and TACRED dataset results show in Table 4.

For -EA, we use the feature vectors of the two entities directly as input to the EDS module. As for -EDS, we remove the global semantic feature vector output from the input of the classifier. The results show that Entity-aware attention has the highest contribution to the F1-score, while the Entity-driven sentence encoding module has a relatively small impact. We suggest that this phenomenon is due to the absence of contextual information about entities from the EA module, which resulted in the loss of global information captured by EDS. As a consequence, relying solely on these features for relationship prediction can confuse the model and significantly reduce its performance. Since EA has already captured sufficient entity context information to determine the relationship between the entities, the loss of global semantic information captured by EDS model has a smaller impact on the model's performance, especially for shorter sentences.

However, incorporating the community detection algorithm into the experimental design has significantly helped both modules to achieve a higher level of performance by facilitating a comprehensive understanding of the sentence as a whole.

5.3. Effectiveness analysis of dependency components

As the length of a sentence increases, the number of potential relations between entities also increases. This leads to a larger search space and more complex decision-making for relation extraction models. However, by leveraging the dependency tree structure of a sentence, models can capture long-distance word-word relations. This allows models to accurately determine relationships between entities, even when they are far apart in the sentence. Our proposed model was tested on entity pairs with varying word distances, and Fig. 5 displays the performance of our best performing CommDet model.

Our model outperformed the baseline in all three groups, and the performance improvement was more pronounced for longer distances, especially when the distance between the entities is larger than 10. These results confirm that our approach is effective at leveraging dependency information and capturing long-distance word-word relations to improve relation extraction task.

5.4. Case study

We present in the Table 5 the attention visualization of CommDet on two examples sampled from Semeval-2010 task 8.

In the first example, our method effectively focuses on the entity context: "Loretta Pettway". This entity is coreferential with "cousin", which is a crucial clue to predict the relationship between the two entities in the sentence. Since the baseline model does not accurately capture this clue, it is not possible to determine the relationship between "cousin" and "quilt" and determine the relationship between them as "Other". We can also observe that the EDS module hardly pays attention to information that is not relevant for predicting the relationship between two entities, which serves as a filter for noise information. This way, the model can focus on the essential features that are critical to predicting the relationship accurately.

As for the second example, we can see that the baseline model predicts the "Contain-container" relationship for the sentence mainly due to the influence of the words "has" and "inside". However, our proposed EDS module successfully captures the crucial phrase "loaded with inside" to form the backbone of the sentence's meaning, while ignoring the influence of noise information, and thereby successfully determining the relationship between the two entities.

6. Analysis and errors

Error analysis is the process of analyzing these mistakes to better understand the model's strengths and weaknesses, and ultimately improve its performance. In this section, we will use the confusion matrix to analyze the errors generated by our model when evaluated

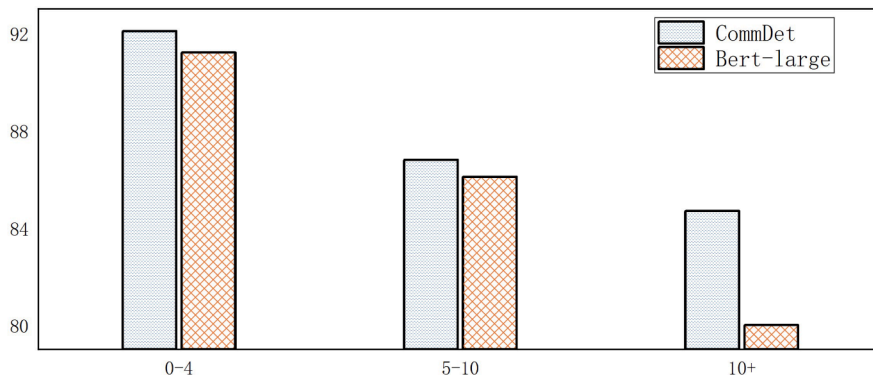


Fig. 5. The performance of BERT-large baseline and CommDet on test instances from SemEval grouped by the entities' distance.

Table 5

Two examples which are sampled from the Semeval-2010 Task 8 test set, with the words shaded to indicate the level of attention allocated to them. Note that words with very low attention values have not been shaded for the purpose of easier perception.

Sampled Examples		Partition results	Prediction
EA	The first exhibition featured seven quilts by Loretta Pettway, Arlonzia Pettway's first cousin. E_1 :quilts E_2 :cousin	S_1 : The first exhibition featured S_2 : Seven quilts	BERT: <i>Other</i>
		S_3 : By Loretta Pettway S_4 : Arlonzia Pettway's first cousin	CommDet: <i>Product-Producer</i>
EDS	The first exhibition featured seven quilts by Loretta Pettway, Arlonzia Pettway's first cousin. E_1 :quilts E_2 :cousin	S_1 : This SNES controller has S_2 : A flash driver	BERT: <i>Contain-container</i>
		S_3 : Loaded with inside S_4 : ROMs	CommDet: <i>Component-Whole</i>
EA	This SENS controller has a flash driver loaded with ROMs inside. E_1 : flash driver E_2 : ROMs	S_1 : This SNES controller has S_2 : A flash driver	BERT: <i>Contain-container</i>
		S_3 : Loaded with inside S_4 : ROMs	CommDet: <i>Component-Whole</i>
EDS	This SENS controller has a flash driver loaded with ROMs inside. E_1 : flash driver E_2 : ROMs	S_1 : This SNES controller has S_2 : A flash driver	BERT: <i>Contain-container</i>
		S_3 : Loaded with inside S_4 : ROMs	CommDet: <i>Component-Whole</i>

on the Semeval test set. The result is shown in Fig. 6.

From the figure, it is evident that the model used in this study misclassifies more sentences as the "Other" category. We argue that this may be because the "Other" category represents relationships that do not fit into any of the pre-defined categories. These relationships may be complex or rare, and the model may not have enough training data to accurately identify them. Consequently, the model may misclassify these relationships as "Other". Moreover, we observed that the model assigns relationships to some entities in sentences whose ground-truth labels are in the "Other" relationship. We believe that this is due to the model's lack of understanding of the semantic information of the entities, resulting in over-reliance on specific phrases that are not indicative of the relationship between the entities. This over-reliance leads to biased or inaccurate results.

7. Conclusion

In this study, we addressed the limitations of existing relation extraction models and proposed a novel approach to identify phrases in sentences by applying community detection algorithms on the corresponding dependency trees. Our experimental results demonstrate the effectiveness of this method in capturing entity context and sentence structure. By using an inheritance structure to capture entity context and sentence structure, we were able to achieve the best performance on two widely used relation extraction datasets, Semeval-2010 task 8 and TACRED.

While constructing the adjacency matrix to be used in the community detection algorithm, we treated the dependency tree's adjacency matrix as an undirected and unweighted graph. However, community detection algorithms can also be applied on weighted graphs. As future work, we plan to assign weights to the edges in the tree based on the type of dependency between the words to produce better partitioning results.

Overall, our proposed approach provides a new perspective on relation extraction and offers promising results. This approach can be further extended by incorporating additional features and improving the performance of the community detection algorithm. Future research could explore the generalizability of our method to other languages and tasks in natural language processing.

CRedit authorship contribution statement

Shuang Liu: Conceptualization. Xunqin Chen: Methodology. Jiana Meng: Supervision. Niko Lukač: Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

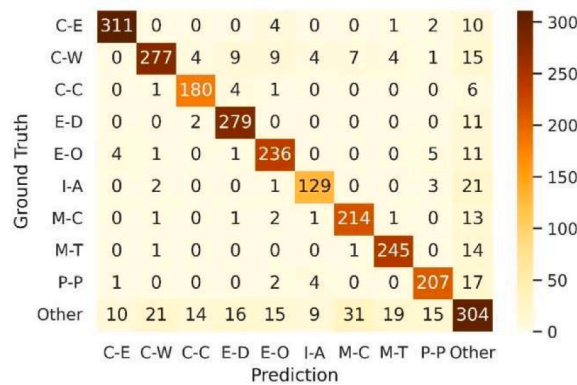


Fig. 6. Confusion matrix.

Data availability

The authors do not have permission to share data.

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References

- Bunescu, R.C., Mooney, R.J., 2005. A shortest path dependency kernel for relation extraction. In: Proceedings of the Conference on Human Language Technology and Empirical Methods in Natural Language Processing - HLT '05. Presented at the the conference. Association for Computational Linguistics, Morristown, NJ, USA. <https://doi.org/10.3115/1220575.1220666>.
- Chen, X., Zhang, N., Xie, X., Deng, S., Yao, Y., Tan, C., Huang, F., Si, L., Chen, H., 2022. KnowPrompt: knowledge-aware prompt-tuning with synergistic optimization for relation extraction. In: Proceedings of the ACM Web Conference 2022. Presented at the WWW '22: The ACM Web Conference 2022. ACM, New York, NY, USA. <https://doi.org/10.1145/3526901.3526902>.
- Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2018. BERT: pre-training of deep bidirectional transformers for language understanding. arXiv [cs.CL].
- Guo, Z., Zhang, Y., Lu, W., 2019. Attention guided graph convolutional networks for relation extraction. arXiv [cs.CL].
- Han, J., Zhao, S., Cheng, B., Ma, S., Lu, W., 2022. Generative prompt tuning for relation classification. arXiv [cs.CL].
- Han, X., Zhao, W., Ding, N., Liu, Z., Sun, M., 2022b. PTR: prompt tuning with rules for text classification. *AI Open* 3, 182–192.
- Huguet Cabot, P.-L., Navigli, R., 2021. REBEL: relation extraction by end-to-end language generation. In: Findings of the Association for Computational Linguistics: EMNLP 2021. Presented at the Findings of the Association for Computational Linguistics: EMNLP 2021. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2021.findings-emnlp.204>.
- Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., Zettlemoyer, L., 2020. BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Presented at the Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2020.acl-main.703>.
- Li, B., Ye, W., Zhang, J., Zhang, S., 2022. Reviewing labels: label graph network with top-k prediction set for relation extraction. arXiv [cs.CL].
- Liang, X., Wu, S., Li, M., Li, Z., 2022. Modeling multi-granularity hierarchical features for relation extraction. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Presented at the Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2022.naacl-main.375>.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V., 2019. RoBERTa: a robustly optimized BERT pretraining approach. arXiv [cs.CL].
- Paolini, G., Athiwaratkun, B., Krone, J., Ma, J., Achille, A., Anubhai, R., Santos, C.N.dos, Xiang, B., Soatto, S., 2021. Structured prediction as translation between augmented natural languages. arXiv [cs.LG].
- Tai, K.S., Socher, R., Manning, C.D., 2015. Improved semantic representations from tree-structured long short-term memory networks. In: Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Presented at the Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.3115/v1/p15-1150>.
- Tian, Y., Chen, G., Song, Y., Wan, X., 2021. Dependency-driven relation extraction with attentive graph convolutional networks. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Presented at the Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2021.acl-long.344>.
- Traag, V.A., Waltman, L., van Eck, N.J., 2019. From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* 9, 5233.
- Wang, L., Cao, Z., de Melo, G., Liu, Z., 2016. Relation classification via multi-level attention CNNs. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Presented at the Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/p16-1123>.
- Xu, K., Reddy, S., Feng, Y., Huang, S., Zhao, D., 2016. Question answering on freebase via relation extraction and textual evidence. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Presented at the Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/p16-1220>.

- Xu, Y., Mou, L., Li, G., Chen, Y., Peng, H., Jin, Z., 2015. Classifying relations via long short term memory networks along shortest dependency paths. In: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing. Presented at the Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/d15-1206>.
- Yamada, I., Asai, A., Shindo, H., Takeda, H., Matsumoto, Y., 2020. LUKE: deep contextualized entity representations with entity-aware self-attention. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP). Presented at the Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2020.emnlp-main.523>.
- Ye, D., Lin, Y., Li, P., Sun, M., 2022a. Packed levitated marker for entity and relation extraction. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Presented at the Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/2022.acl-long.337>.
- Ye, H., Zhang, N., Chen, Hui, Chen, Huajun, 2022. Generative knowledge graph construction: a review. arXiv [cs.CL].
- Zelenko, D., Aone, C., Richardella, A., 2002. Kernel methods for relation extraction. In: Proceedings of the ACL-02 Conference on Empirical Methods in Natural Language Processing - EMNLP '02. Presented at the the ACL-02 conference. Association for Computational Linguistics, Morristown, NJ, USA. <https://doi.org/10.3115/1118693.1118703>.
- Zhang, Y., Qi, P., Manning, C.D., 2018. Graph convolution over pruned dependency trees improves relation extraction. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Presented at the Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/d18-1244>.
- Zhang, Y., Zhong, V., Chen, D., Angeli, G., Manning, C.D., 2017. Position-aware attention and supervised data improve slot filling. In: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing. Presented at the Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/d17-1004>.
- Zheng, S., Wang, F., Bao, H., Hao, Y., Zhou, P., Xu, B., 2017. Joint extraction of entities and relations based on a novel tagging scheme. In: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Presented at the Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, Stroudsburg, PA, USA. <https://doi.org/10.18653/v1/p17-1113>.
- Zhou, W., Chen, M., 2021. An improved baseline for sentence-level relation extraction. arXiv [cs.CL].